



CORE PROJECT DOCUMENTATION PACK

Technical & Product Architecture

Document 04 of 08 · PRODUCT & AI

M A D HALO Technologies



This document explains, in plain terms, how HI NGE SIYANI is put together and why each part exists. It's written to be understandable by non-technical partners and advisors, not just developers.

Overview

HI NGE SIYANI has three broad layers:

1. **The application** — what the user sees and interacts with (Delphi 12 FMX)
2. **The backend** — where accounts, data, and security live (Supabase)
3. **The intelligence layer** — how Siyani understands and responds to users (AI/LLM, knowledge, and multilingual layers)

Delphi 12 FMX Application

This is the user-facing application — the actual interface people use on their device. Delphi 12 FMX (FireMonkey) was chosen because it allows building a single application that can run across multiple platforms (desktop and mobile) from one codebase. This layer is responsible for onboarding, language selection, the chat/interaction experience with Siyani, and displaying profile and history information.

Supabase Authentication

Handles user sign-up, login, and identity — confirming that a user is who they say they are, and keeping their account secure. This is what allows a user's profile, language preference, and conversation history to be tied to them personally and kept private.

Supabase Database

Stores the structured data the application needs to function: user profiles, language preferences, conversation history, and (over time) curated content used to support Siyani's answers. Think of this as the ecosystem's memory — everything that needs to persist between sessions lives here.

Supabase Security / Row-Level Security (RLS)

Row-Level Security is a set of rules that control exactly who can see or change which pieces of data. In practice, this means a user's profile, preferences, and conversation history are only accessible to that user (and to authorised system processes) — not visible to other users or exposed by mistake. This is a foundational trust and privacy safeguard, not an optional extra.

AI/LLM Layer

This is where Siyani's actual reasoning happens — the layer that takes a user's question and produces a helpful, accurate response. At this stage, this layer is built on top of an existing large language model, guided by carefully



designed prompts and, increasingly, by the curated content in the knowledge layer below (see the Siyani AI Assistant Strategy document for more detail on this approach).

Knowledge/Content Layer

This is the curated, verified information Siyani draws on when answering education and public-service questions — as opposed to relying purely on the AI model's general background knowledge. Over time, this layer becomes the project's most valuable asset: a trusted, maintained body of accurate, locally relevant content.

Multilingual Layer

This layer manages how Siyani understands input in a user's chosen language and produces natural, accurate output in that same language. It sits between the AI/LLM layer and the user, ensuring that language quality (not just literal translation) is treated as a first-class concern.

Future Speech-to-Text / Text-to-Speech Layer

Planned for a later phase, this layer will convert spoken input into text Siyani can process (speech-to-text) and convert Siyani's responses into natural spoken output (text-to-speech), across the ecosystem's supported languages. This is not yet built.

User Profiles and Preferences

Stores each user's identity-linked information: language preference, relevant profile details (e.g. education stage, areas of interest), and settings. This is what allows Siyani's responses to be personalised rather than generic.

Conversation/History

Keeps a record of a user's past interactions with Siyani, so context isn't lost between sessions and so users can revisit previous guidance. This also supports continuity when a user switches languages mid-conversation.

Future Integrations

Planned, not yet built: potential connections to external systems such as education institutions' information sources, official government/public-service data sources, and possibly community or partner organisation systems. These would extend what Siyani can reliably help with, without requiring HI NGE SIYANI to manually maintain every piece of information itself.

How It Fits Together (Plain-Language Summary)

A user opens the app (Delphi 12 FMX), signs in securely (Supabase Auth), and their data is protected so only they can see it (Supabase RLS). They pick a language, and from then on, Siyani (AI/LLM layer, supported by the multilingual and knowledge layers) understands and responds to them in that language — drawing on curated,



verified content wherever possible rather than guessing. Everything about their profile and conversation is saved (Supabase database) so the experience is continuous over time. Voice and external integrations are the next major additions once the current core is solid.